# Games that Teach Project Plan

## Team Members

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## Notes / Ideas

- Actual gameplay loop vs. integration with incremental style - maximizing score as an educational method?
- Final score has to be calculated by the player
- Deck builder style game?
- Start with a base number, arrange effects to maximize score
- Positional effect cards that rely on their order relative to the other effect cards
- Effects/bonuses based on properties of base number
- Card/number property filter (include/exclude primes, multiples of certain numbers, etc)
- Increasing minimum score to continue progressing

## Game Details

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|---|---|
| **Game Title** | Smooth Operator |
| **Game Concept** | The player is trying to maximize a score with random cards they collect by placing them in the most efficient order, the order of operations is only based on cards.<br><br>The player is trying to maximize their score by performing arithmetic operations on a random number in a certain order<br><br>The game ends when the player fails to reach a minimum score |

# Educational Goals

## Learning Objective

After playing the game, the player should be better able to quickly perform mathematical calculations. 6.NS.B. Compute fluently with multi-digit numbers and find common factors and multiples.

## Practice

The player improves this skill by doing math with operations in their head in the game.

## Feedback

The player can judge their score relative to the required amount and gauge if they are ahead or behind.

# Target Audience

## Age Range / Grade Level

The target grade range is 6th+

## Background Knowledge

Students need to know basic multiplication, division, subtraction, and addition

## Considerations

They might get bored quickly while playing the game or find the game confusing, we can deal with this by ensuring the game is engaging and also including a basic tutorial.

# Core Gameplay

## Gameplay Cycle

Given a base number and a fixed amount of modifiers/operations → enable, disable, and re-order the operations → selected operations are applied in the specified order on the base number → the result is added to the score → separate, user-curated score effects (if they have been acquired) are applied to the player's score → if the player's score is above a minimum, they are shown to the shop, where they are granted a choice between three score effects → repeat

Start round:
Number: 23
Preset cards:

- 2x
- -4
- /3

And then you could put your cards in between to get something like,

- 2x
- If the number is even, 3x
- -4
- /3

And then once you're satisfied you would run it and it would total your score.

## Player Controls

| Player Action | Control | What Happens? |
|---|---|---|
| Card movement | Mouse | Allows you to move cards around by dragging them |
| | | |

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# Game Mechanics

## Movement

They interact by choosing where they place cards using their mouse cursor

## Scoring/Progress

They can tell they are succeeding by looking at their score relative to the minimum score needed

## Challenge/Obstacles

The player has to consider the order they should do their operations in to maximize their scare

## Consequences

They end up falling behind in score and losing if they made a large mistake

## Increasing Difficulty

Likelihood of limiting elements of cards increases with rounds played

## Educational Mechanic

Arranging cards in such a way to produce the maximum possible score.

# Gameplay

The game play has the following general feel at the beginning points.

| | |
|---|---|
| **Beginning** | |
| **Beginning** | |

| | |
|---|---|
| **End** | |

# Look/Feel

## Setting / Theme

N/A, No narrative focus and highly abstract

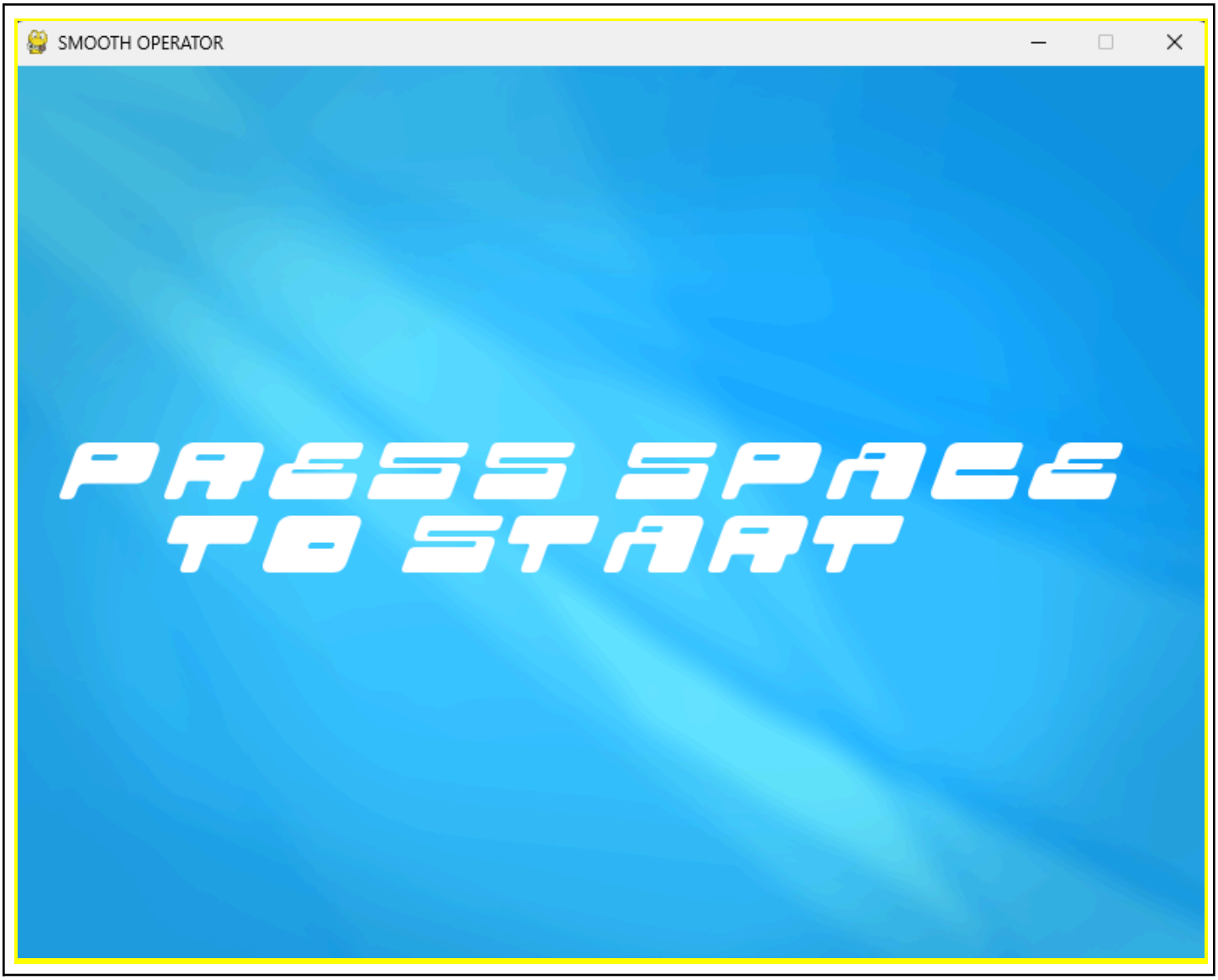
## Visual Style

Abstract skeuomorphic style inspired by Windows Aero and early 3D CGI

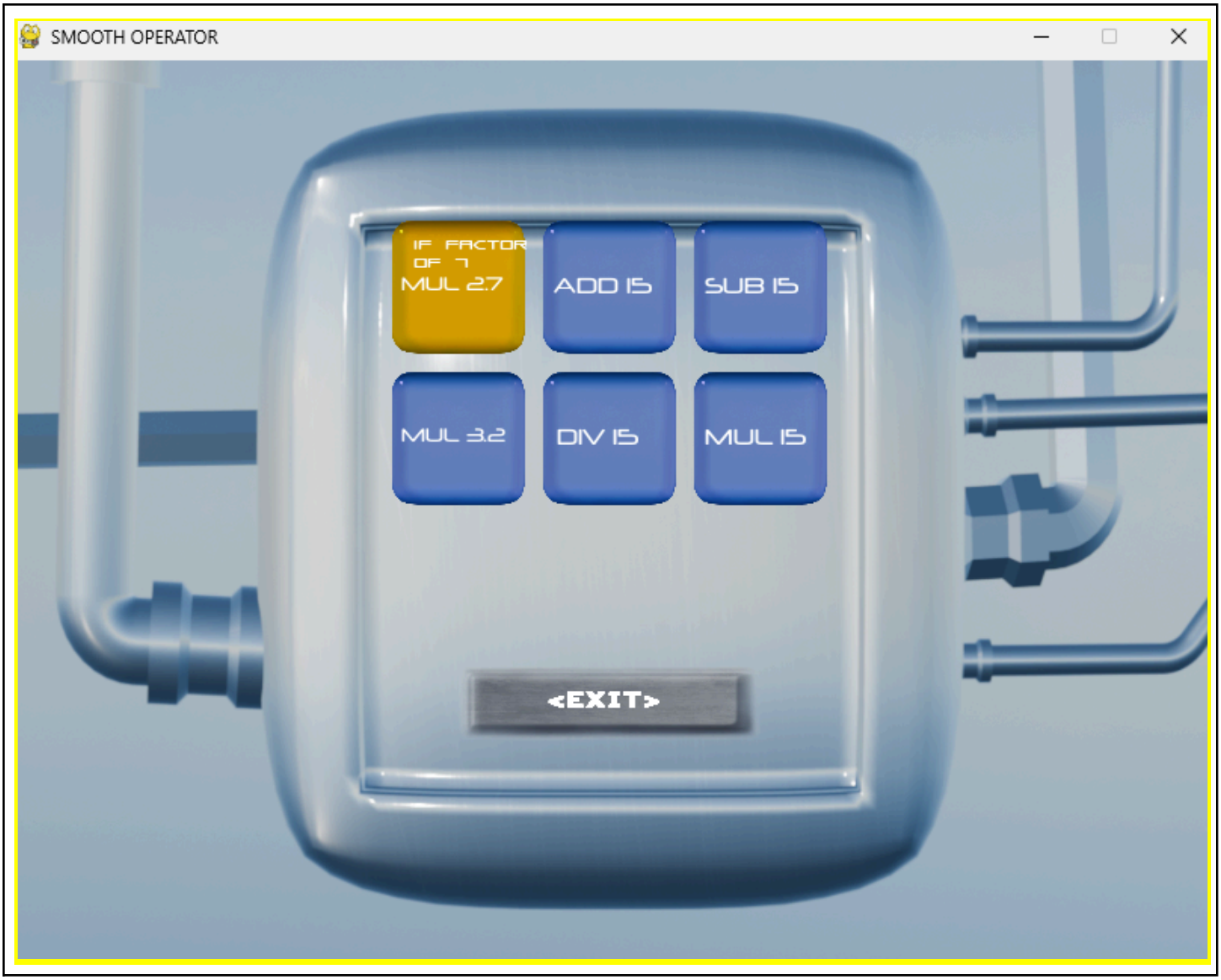
## Mood

The player should feel engaged and rewarded for their effort, but still be forced to consider their choices carefully and keep the relevant math in mind.

# Visuals


SMOOTH OPERATOR
PRESS SPACE
TO START


SMOOTH OPERATOR
3851
MUL 15
MUL 15
ADD 80
IF FACTOR OF 7
MUL 18.2
79
IF IN SLOT 2
SUB 9.3
SUB 11




# Revisions

| Revision Description | How It Improved the Experience |
|---|---|
| Remove bonus scoring mechanic | Limited point gain in order to prolong the sense of meaningful rewards |
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